

COMPUTER SCIENCE B.S. DEGREE REQUIREMENTS

University: TechState University

Catalog Year: 2024–2025

Department: Electrical Engineering and Computer Science (EECS)

Document ID: CAT-PROG-001

Source: <https://catalog.techstate.edu/programs/cs-bs-2024>

Date Accessed: March 29, 2025

1. OVERVIEW

The Bachelor of Science in Computer Science requires a minimum of 128 credit hours. Students must complete all required core courses, a designated concentration, a senior capstone, and a set of electives. Students must maintain a minimum cumulative GPA of 2.0 and a minimum CS GPA of 2.0 to remain in good standing.

2. GENERAL EDUCATION REQUIREMENTS (30 credits)

All CS students must complete TechState's General Education requirements:

- Writing I (ENG 101, 3 credits) — must be completed in first year
- Writing II (ENG 102, 3 credits) — prerequisite: ENG 101
- Quantitative Reasoning (MATH 151 or higher, 3 credits)
- Natural Science with lab (2 courses, 8 credits)
- Social Science (2 courses, 6 credits)
- Humanities (2 courses, 6 credits)
- Global Perspectives (1 course, 3 credits)

3. MATHEMATICS REQUIREMENTS (22 credits)

All CS majors must complete:

- MATH 151: Calculus I (4 credits) — no prerequisite
- MATH 152: Calculus II (4 credits) — prerequisite: MATH 151
- MATH 231: Discrete Mathematics (3 credits) — prerequisite: MATH 151 (may be taken concurrently)
- MATH 241: Linear Algebra (3 credits) — prerequisite: MATH 152
- MATH 251: Probability and Statistics for CS (3 credits) — prerequisite: MATH 152; co-requisite: CS 201
- MATH 301: Differential Equations (3 credits) — prerequisite: MATH 152 — Required for AI/ML concentration only; optional for others

4. CORE CS COURSES (43 credits)

The following courses are required for all CS majors regardless of concentration:

- CS 101: Introduction to Programming (3 credits) — no prerequisite
- CS 102: Data Structures and Algorithms I (3 credits) — prerequisite: CS 101; minimum grade C
- CS 201: Data Structures and Algorithms II (4 credits) — prerequisite: CS 102; minimum grade C

- CS 210: Computer Organization and Architecture (3 credits) — prerequisite: CS 101; co-requisite: CS 102
- CS 220: Operating Systems (3 credits) — prerequisite: CS 201 and CS 210; minimum grade C
- CS 230: Computer Networks (3 credits) — prerequisite: CS 201
- CS 240: Software Engineering (3 credits) — prerequisite: CS 201
- CS 250: Theory of Computation (3 credits) — prerequisite: CS 201 and MATH 231
- CS 260: Database Systems (3 credits) — prerequisite: CS 201; minimum grade C-
- CS 301: Algorithms (4 credits) — prerequisite: CS 201 and MATH 231; minimum grade C
- CS 310: Programming Languages (3 credits) — prerequisite: CS 201 and CS 250
- CS 320: Systems Programming (3 credits) — prerequisite: CS 210 and CS 220
- CS 499: Senior Capstone Project (4 credits) — prerequisite: Senior standing (90+ credits completed), CS 301, and completion of at least one concentration course; minimum grade C

5. CONCENTRATION REQUIREMENTS (choose one, 15–18 credits)

Students must complete one of the following concentrations:

5a. Artificial Intelligence and Machine Learning (18 credits)

- CS 351: Machine Learning Foundations (3 credits) — prerequisite: CS 301 and MATH 241 and MATH 251
- CS 352: Deep Learning (3 credits) — prerequisite: CS 351
- CS 353: Natural Language Processing (3 credits) — prerequisite: CS 351
- CS 354: Computer Vision (3 credits) — prerequisite: CS 351
- CS 355: Reinforcement Learning (3 credits) — prerequisite: CS 352
- CS 356: AI Ethics and Fairness (3 credits) — prerequisite: CS 351; may be taken concurrently with CS 352

Choose 4 of the 6 courses above (12 credits from CS 351 + any 3 electives); CS 351 is mandatory.

5b. Systems and Security (15 credits)

- CS 361: Cybersecurity Fundamentals (3 credits) — prerequisite: CS 220 and CS 230
- CS 362: Cryptography (3 credits) — prerequisite: CS 250 and MATH 231
- CS 363: Distributed Systems (3 credits) — prerequisite: CS 220 and CS 230
- CS 364: Cloud Computing (3 credits) — prerequisite: CS 363; co-requisite allowed: CS 361
- CS 365: Penetration Testing (3 credits) — prerequisite: CS 361; minimum grade B-; instructor consent required

Choose all 5 courses.

5c. Software Engineering (15 credits)

- CS 371: Advanced Software Engineering (3 credits) — prerequisite: CS 240 and CS 260
- CS 372: Human-Computer Interaction (3 credits) — prerequisite: CS 240
- CS 373: Mobile Application Development (3 credits) — prerequisite: CS 240
- CS 374: Web Technologies (3 credits) — prerequisite: CS 240
- CS 375: Software Quality and Testing (3 credits) — prerequisite: CS 371

Choose all 5 courses.

6. FREE ELECTIVES (13–16 credits)

Students may take any courses at the 200-level or above to satisfy the elective requirement, including CS courses from outside their concentration, MATH courses beyond requirements, or courses from other departments with advisor approval. At least 6 credits of electives must be CS courses numbered 300 or above.

7. RESIDENCY REQUIREMENT

At least 30 of the 128 credits required for the degree must be completed at TechState (cannot be transferred). Of those 30, at least 18 must be CS courses numbered 300 or above.

8. GRADUATION CHECK

Students must apply for graduation by the end of the semester prior to their expected graduation semester. A final audit is conducted by the registrar. Students with outstanding incompletes (grade of I) in required courses cannot graduate until the incomplete is resolved.

9. ACADEMIC STANDING

Students who earn below a C in two or more required core CS courses in the same semester may be placed on academic probation. Students on probation may not register for more than 15 credit hours per semester. Dismissal from the CS program may result from three consecutive semesters on probation.